

# **3D Finite-Element Groundwater Flow Modelling of the Semi-Arid Gundlu Sub-Basin Using FEFLOW**

Major Project –II (CV499) Report  
submitted in partial fulfilment of the requirement for the degree of

**BACHELOR OF TECHNOLOGY**

In

**CIVIL ENGINEERING**

By

**Gowtham R (221CV222)**

**Kandolu Vasantha Raju (221CV225)**

Under the guidance of

**Dr. Dwarakish G S**



**DEPARTMENT OF CIVIL ENGINEERING  
NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA  
SURATHKAL, MANGALORE-575025**

**MAY 2026**

## DECLARATION

We hereby declare that the report of the Major project-II (CV499) entitled “**3D Finite-Element Groundwater Flow Modelling of the Semi-Arid Gundlu Sub-Basin Using FEFLOW**”, which is being submitted to **National Institute of Technology Karnataka, Surathkal** in partial fulfilment of requirements for the award of the degree of Bachelor of Technology in Civil Engineering is a Bonafide record of the work carried out by us. The material contained in this report has not been submitted to any university or Institution for the award of any degree

1. Gowtham R (221CV222),
2. Kandolu Vasantha Raju (221CV225)

Place: NITK, Surathkal

Date: May 2026

## **CERTIFICATE**

This is to certify that this report entitled “**3D Finite-Element Groundwater Flow Modelling of the Semi-Arid Gundlu Sub-Basin Using FEFLOW**” submitted by,

- 1. GOWTHAM R (221CV222),**
- 2. KANDOLU VASANTHA RAJU (221CV225)**

as the record of the work carried out by them in the Department of Civil Engineering, is accepted as the B-Tech Major Project –II Report (CV499) submission in partial fulfilment of the requirement for the award degree of **Bachelor of Technology in Civil Engineering**.

Dr. Dwarakish G S  
Professor & Guide  
Department of Civil Engineering  
NITK, Surathkal

Dr Basavaraju Manu  
Chairperson – DUGC  
Department of Civil Engineering  
NITK, Surathkal

## **ACKNOWLEDGEMENTS**

We would like to express our sincere gratitude to R Avinash for their valuable guidance, support and encouragement throughout the course of this project. Their insights and suggestions significantly contributed to the successful completion of this work. We are also thankful for their continuous assistance during the experimental work, data collection and overall development of the study. Their support helped us overcome various challenges encountered during the project. We would also like to extend our appreciation to all those who directly or indirectly contributed to this work.

## LIST OF TABLES

|                                                                          |     |
|--------------------------------------------------------------------------|-----|
| DECLARATION .....                                                        | i   |
| CERTIFICATE .....                                                        | ii  |
| ACKNOWLEDGEMENTS .....                                                   | iii |
| LIST OF FIGURES .....                                                    | vi  |
| LIST OF TABLES .....                                                     | vi  |
| ABSTRACT .....                                                           | 1   |
| CHAPTER 1: INTRODUCTION .....                                            | 2   |
| 1.1 Background .....                                                     | 2   |
| 1.2 Hard-Rock Aquifers: Concept and Significance.....                    | 2   |
| 1.3 Importance of Groundwater in Agriculture .....                       | 3   |
| 1.4 Problem Statement .....                                              | 3   |
| 1.5 Objectives .....                                                     | 4   |
| CHAPTER 2: LITERATURE REVIEW .....                                       | 5   |
| 2.1 Overview of Groundwater Studies in Hard-Rock Terrains .....          | 5   |
| 2.2 Supporting Literature on Fracture-Based Groundwater Modelling.....   | 5   |
| 2.3 Aquifer Characteristics in South Indian Crystalline Formations ..... | 6   |
| 2.4 FEFLOW and MODFLOW Applications in Hard-Rock Modelling .....         | 7   |
| 2.5 Groundwater Depletion and Over-Extraction in Semi-Arid India.....    | 7   |
| CHAPTER 3: STUDY AREA .....                                              | 8   |
| 3.1 Location and Geographic Setting.....                                 | 8   |
| 3.2 Climate and Rainfall .....                                           | 9   |
| 3.3 Geological and Lithological Characteristics .....                    | 9   |
| 3.4 Hydrogeology of the Study Area .....                                 | 9   |
| 3.5 Borewell Development and Aquifer Parameters .....                    | 10  |
| 3.6 Land Use and Recharge Characteristics .....                          | 10  |
| CHAPTER 4: DATA AND METHODOLOGY .....                                    | 11  |

|                                                            |    |
|------------------------------------------------------------|----|
| 4.1 Data Sources .....                                     | 11 |
| 4.2 FEFLOW Model Setup .....                               | 12 |
| 4.2.1 Mesh Generation .....                                | 12 |
| 4.2.2 Vertical Discretisation: 3D Layer Configuration..... | 12 |
| 4.2.3 Initial and Boundary Conditions .....                | 14 |
| 4.2.4 Parameter Assignment .....                           | 15 |
| 4.2.5 Simulation Settings .....                            | 15 |
| CHAPTER 5: RESULTS AND DISCUSSION.....                     | 16 |
| 5.1 Simulated Hydraulic Head Distribution.....             | 16 |
| 5.2 Simulated versus Observed Head Comparison.....         | 17 |
| 5.3 Model Performance Metrics .....                        | 18 |
| 5.4 Residual Analysis.....                                 | 19 |
| 5.5 Diagnosis and Calibration Recommendations .....        | 20 |
| CHAPTER 6: CONCLUSIONS .....                               | 21 |
| REFERENCES .....                                           | 22 |

## LIST OF FIGURES

|                                                                                                      |    |
|------------------------------------------------------------------------------------------------------|----|
| Figure 1 Aquifer system in fractured rock formations (Modified from Freeze and Cherry, 1979)         | 3  |
| Figure 2 Gundlu Sub-Basin Location map showing basin boundary, river network and observation wells   | 8  |
| Figure 3 FEFLOW modelling workflow flowchart – Gundlu Basin (Source: FEFLOW Workflow document, NITK) | 13 |
| Figure 4 Soil surface distribution and FEFLOW model with assigned hydraulic parameters               | 14 |
| Figure 5 3D visuals of assigned hydraulic conductivity and hydraulic head parameters in FEFLOW       | 15 |
| Figure 6 Simulated vs observed hydraulic head time series for selected wells – FEFLOW Run 4          | 16 |
| Figure 7 Scatter plot: Simulated vs Observed hydraulic head at 38 observation wells – 1:1 line shown | 17 |
| Figure 8 Model performance per well – Gundlu Basin FEFLOW (absolute error and status classification) | 18 |
| Figure 9 Residual plot (Simulated – Observed head) over time and per well – FEFLOW Run 4             | 19 |

## LIST OF TABLES

|                                                                                 |    |
|---------------------------------------------------------------------------------|----|
| Table 1 FEFLOW hydraulic parameters by layer (GEC-2015, Eastern Dharwar Craton) | 10 |
| Table 2 Datasets used in the FEFLOW groundwater model                           | 11 |
| Table 3 Model performance statistics – FEFLOW Run 4 (pre-calibration)           | 18 |

## ABSTRACT

This study presents the development of a 3D finite-element groundwater flow model for the semi-arid Gundlu Sub-Basin in Gundlupet Taluk, Karnataka, using FEFLOW 10.0. The research focuses on understanding groundwater behaviour in hard-rock crystalline aquifers where groundwater occurrence is controlled mainly by fractures, joints, and weathered zones due to low primary porosity. The model integrates DEM data, watershed boundaries, stream networks, borewell information, recharge estimates, and hydraulic parameters derived from GEC-2015 norms and CGWB datasets. A three-layer, four-slice transient groundwater model was developed to simulate hydraulic head distribution and groundwater flow over a five-year period. Simulation results indicate that the model successfully captures regional groundwater flow trends and seasonal fluctuations, with simulated hydraulic heads showing reasonable agreement with observed CGWB well data. Statistical evaluation using RMSE, MAE, and  $R^2$  revealed that the model performance improves significantly after excluding a few boundary-related outlier wells. Residual analysis identified systematic under-prediction caused by recharge and boundary condition uncertainties, highlighting the need for further calibration. The study demonstrates the applicability of FEFLOW for modelling complex hard-rock aquifer systems and provides a scientific basis for sustainable groundwater management and future groundwater depletion assessment in the Gundlu Basin.



## INTRODUCTION

### 1.1 Background

Groundwater is a vital natural resource essential for sustaining life, ecosystems, and agricultural productivity across India. Among the two major types of freshwater sources surface water and groundwater the latter plays a particularly critical role in regions where rainfall is erratic and surface storage is insufficient. In India, groundwater serves as the primary source of irrigation and domestic water supply, especially during dry seasons or drought conditions. In hard-rock regions such as southern Karnataka, groundwater does not occur uniformly within porous sedimentary layers. Instead, it is stored and transmitted through secondary features fractures, joints, shear zones, and weathered horizons. This makes the hydrogeology inherently heterogeneous, spatially variable, and difficult to characterise without detailed subsurface data and numerical modelling.

The Gundlu Basin (Gundlupet Taluk, Chamarajanagar District) exemplifies the stress facing semi-arid hard-rock aquifer systems. Intensive agricultural extraction, episodic monsoon recharge, and low primary porosity create a condition where borewell water tables decline rapidly during dry periods. Advanced 3D numerical modelling using FEFLOW a finite-element groundwater flow simulator offers the most rigorous approach to simulate, understand, and predict this behaviour.

### 1.2 Hard-Rock Aquifers: Concept and Significance

Hard-rock aquifers differ fundamentally from sedimentary aquifers. Primary porosity in crystalline formations such as granite, gneiss, and charnockite is negligible ( $< 1\%$ ). Groundwater occurrence depends entirely on secondary porosity: the network of fractures, joints, and the overlying saprolitic (weathered) zone. The conceptual model of such aquifers, originally described by Freeze and Cherry (1979), identifies a two-component system:

A shallow, porous weathered zone (saprolite), 5–15 m thick, with moderate specific yield ( $S_y \approx 0.03$ ) that acts as the primary storage reservoir.

A deeper fractured bedrock zone, with low transmissivity but high depth extent (–60–120 m), where fractures serve as conduits for groundwater flow.

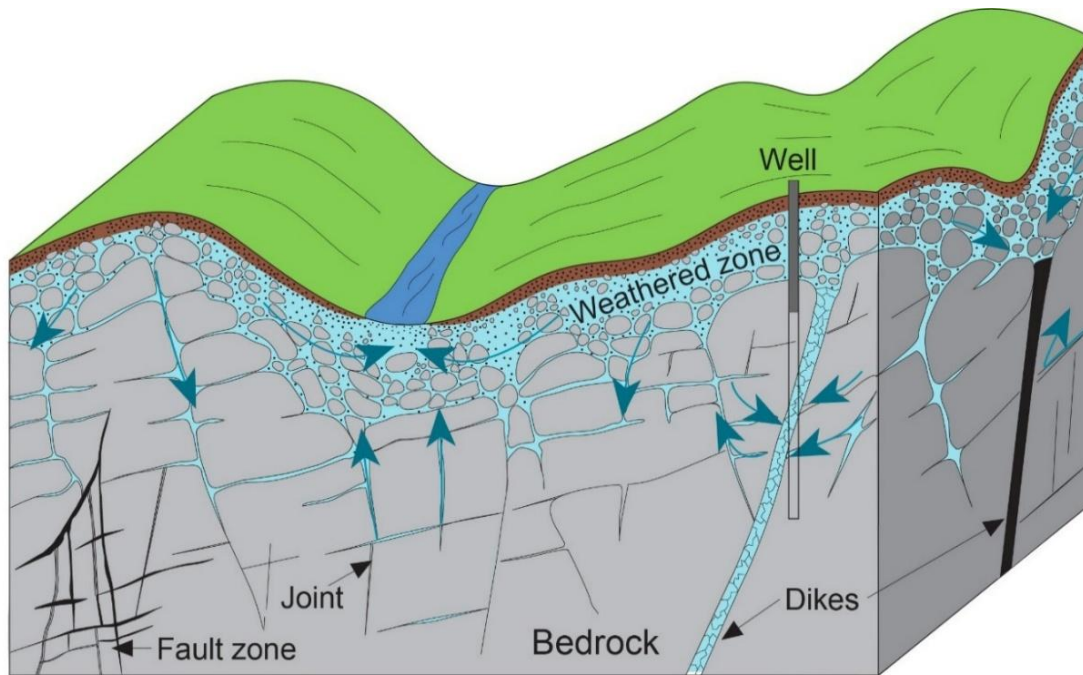


Figure 1 Aquifer system in fractured rock formations (Modified from Freeze and Cherry, 1979)

### 1.3 Importance of Groundwater in Agriculture

Groundwater is the backbone of Indian agriculture, accounting for more than 60% of irrigation water nationally (Wada et al., 2010). It ensures year-round crop production and supports multiple cropping systems by providing water during non-monsoon periods. In hard-rock regions such as the Gundlu Basin, where surface water is limited to seasonal flow in the Gundlu River (a tributary of the Kabini), groundwater through borewells enables irrigation of ragi, maize, pulses, and sugarcane throughout the year.

### 1.4 Problem Statement

The Gundlu Basin is experiencing accelerating groundwater depletion driven by a combination of anthropogenic and geological factors. High borewell density for agricultural irrigation continues to intensify without corresponding managed aquifer recharge, while the hard-rock geology of the Eastern Dharwar Craton inherently limits recharge efficiency. Compounding this, climate variability is reducing effective monsoon infiltration, further constraining natural replenishment of the system. Critically, the absence of a calibrated 3D numerical groundwater model has left the basin-scale water

budget unquantified, preventing informed management intervention. These factors collectively underline the urgent need for a scientifically rigorous 3D finite-element model of the basin, which this study develops using FEFLOW 10.0.

## **1.5 Objectives**

This study is guided by a set of interconnected objectives aimed at advancing both the scientific understanding and practical management of groundwater in the Gundlu Basin. The study first seeks to understand groundwater importance, depletion dynamics, and hard-rock hydrogeology in the semi-arid basin context, before developing a precise 3D finite-element groundwater model using FEFLOW to represent the weathered and fractured subsurface layers. Spatially distributed hydraulic parameters including  $K_{xx}$ ,  $K_{yy}$ ,  $K_{zz}$ ,  $S_y$ , and  $S_s$  are assigned based on GEC-2015 norms and local borewell data to ensure physical representativeness. Transient hydraulic head distribution is then simulated over a multi-year period and compared against CGWB observation well data. Model performance is assessed using statistical metrics including RMSE, MAE, and  $R^2$ , with calibration priorities identified for subsequent runs. Ultimately, the study aims to support sustainable groundwater management planning for the Gundlupet Taluk by providing a robust quantitative modelling framework.

### LITERATURE REVIEW

#### 2.1 Overview of Groundwater Studies in Hard-Rock Terrains

Groundwater occurrence in hard-rock regions has been extensively studied owing to the discontinuous and fracture-controlled nature of aquifers. Hard-rock aquifers (granite, gneiss, charnockite) lack primary porosity and depend entirely on secondary porosity fractures, joints, faults, and weathered zones. Studies by Maréchal et al. (2020) and Dewandel et al. emphasise that groundwater movement in these terrains is principally through three layers: the weathered zone (saprolite), the shallow fractured zone, and the deeper fracture network. This three-layer conceptual model aligns strongly with the hydrogeology of the Gundlupet–Gundlu Basin, where borewell drilling commonly intersects multiple fractures at different depths. CGWB (2012) and Atal Bhujal Yojana (2023–24) further confirmed that the Gundlupet block operates within a low-storage, hard-rock aquifer system exhibiting rapid water level decline under intensive pumping.

#### 2.2 Supporting Literature on Fracture-Based Groundwater Modelling

Regional and case-study literature consistently reinforces the significance of fracture characterisation and numerical modelling in hard-rock aquifer systems. BHP Billiton (2009) demonstrated through the Olympic Dam Expansion Draft EIS that regional-scale groundwater models can effectively evaluate deep aquifer systems, with mesh-based numerical frameworks capable of handling localised hydraulic stresses across complex geological settings. This finding is directly relevant to the Gundlu Basin, where localised fracture-induced stresses govern yield variability across borewells.

Ramstad et al. (2010), in a study on ground source energy systems in crystalline bedrock, highlighted that explicit fracture mapping is essential prior to modelling, as fracturing fundamentally alters the geometry of the flow network. Their work emphasised that the overall yield of a crystalline aquifer system is controlled by fracture network connectivity rather than matrix permeability alone a conclusion consistent with the fracture-dominated hydrogeology of the Gundlupet–Gundlu Basin.

Ma et al. (2016), in a simulation study on groundwater level recovery, integrated pipe-flow models to represent fractures as discrete conduits rather than equivalent porous media. Their Discrete Fracture Element (DFE) approach predicted recovery patterns with high accuracy,

underscoring the value of fracture-explicit modelling where borewell data captures multiple fracture intersections at depth.

Yang et al. (2010) contributed a data model framework for system conceptualisation in fractured-rock settings, demonstrating how conceptual geological frameworks can be systematically translated into numerical grids. Their workflow provides a clear methodological basis for converting borewell lithology logs and fracture-depth records into spatially distributed model layers, directly applicable to the 3D stratigraphic modelling approach adopted for the Gundlu Basin.

Varalakshmi et al. (2014) developed a two-layer hard-rock aquifer model for a crystalline terrain case study and showed that intensive groundwater draft causes severe water level decline. Critically, their simulations indicated that a 40% reduction in pumping draft is sufficient to initiate recovery of groundwater levels, offering a quantitative management benchmark applicable to similarly stressed basins like Gundlupet.

More recently, Pareta (2024) demonstrated in a groundwater contamination modelling study of the Ayad River Basin that complex geometric subsurface data can be successfully integrated with GIS inputs to achieve high-accuracy numerical modelling outcomes. This confirms that GIS-derived spatial datasets including borewell coordinates, fracture depths, and basin boundaries can serve as reliable inputs for constructing and calibrating numerical groundwater models in data-limited hard-rock regions.

## **2.3 Aquifer Characteristics in South Indian Crystalline Formations**

Several studies conducted across Karnataka, Tamil Nadu, and Andhra Pradesh highlight that crystalline aquifers show weathered zone thickness of 5–15 m, first fracture horizons at 25–40 m bgl, second fracture horizons at 50–80 m bgl, and deep fracture zones at 90–120 m bgl. Research from the Berambadi Catchment (adjacent to Gundlupet) shows similar fracture patterns and recharge behaviours. CGWB (2012) and Atal Bhujal Yojana (2023–24) confirm that the Gundlupet block groundwater system is characterised by low storage and rapid decline under intensive pumping identical to the conditions modelled in this study.

## **2.4 FEFLOW and MODFLOW Applications in Hard-Rock Modelling**

FEFLOW (DHI WASY), a 3D finite-element subsurface flow simulator, has been used extensively for hard-rock and fractured aquifer systems owing to its ability to handle unstructured meshes, variable-density flow, and discrete fracture elements (DFEs). It outperforms finite-difference codes such as MODFLOW in capturing complex geometries arising from irregular geological contacts, basin boundaries, and fracture networks. Studies by *Frontiers in Water* (2022) and *Nature* (2022) indicate that recharge in the Gundlupet region is episodic, occurring mainly during high-intensity rainfall events, requiring time-varying recharge boundary conditions a feature well handled by FEFLOW's time-series import capability.

## **2.5 Groundwater Depletion and Over-Extraction in Semi-Arid India**

India is the largest user of groundwater globally, relying on borewells for over 60% of irrigation water (Wada et al., 2010). In Karnataka, particularly Mysuru and Chamarajanagar Districts, high borewell dependency combined with low recharge and crystalline rock results in progressive water table decline. The Gundlu Basin exhibits these same patterns, making it a critical case study for 3D groundwater modelling and sustainability assessment.

### 3.1 Location and Geographic Setting

The study area, the Gundlu Sub-Basin, is located in Gundlupet Taluk in the southern part of Karnataka, within the Mysuru–Chamarajanagar region. It forms part of the Kabini River sub-basin, which ultimately drains into the Cauvery River system. The basin lies within a semi-arid climatic zone characterised by moderate and unevenly distributed monsoon rainfall and high evapotranspiration rates. The topography is gently undulating, with elevations ranging from approximately 750 to 900 m above mean sea level (m ASL).

**Basin area:** approximately 1,270 km<sup>2</sup>

**Average annual rainfall:** approximately 600 mm

**Main water body:** Gundlu River (tributary of Kabini)

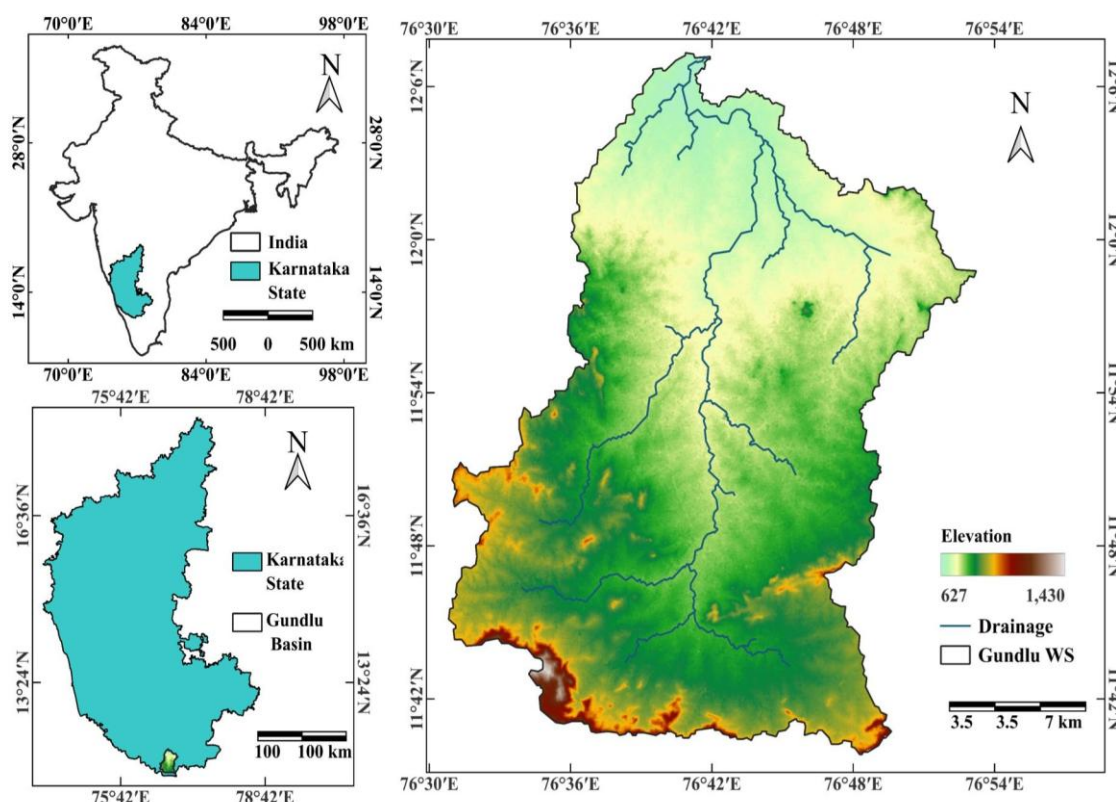


Figure 2 Gundlu Sub-Basin Location map showing basin boundary, river network and observation wells

### 3.2 Climate and Rainfall

The study area experiences a tropical semi-arid climate with three major seasons: summer (March–May, temperatures 32–36°C), monsoon (June–October, the major rainfall period), and winter (November–February). Average annual rainfall ranges from 650–800 mm, predominantly received during the southwest monsoon. Intense short-duration rainfall events contribute significantly to episodic recharge, a key characteristic of hard-rock aquifer systems in southern India (Frontiers in Water, 2022).

### 3.3 Geological and Lithological Characteristics

The Gundlu Basin is underlain by Precambrian crystalline hard rocks of the Eastern Dharwar Craton, predominantly:

Granite gneiss (GR02/GN02 GEC-2015 classification)

Hornblende-biotite gneiss

Charnockite (CK01 GEC-2015 classification)

Granulite

Dolerite dykes and minor quartz veins

These rocks have very low primary porosity, and groundwater is stored mainly in the weathered (saprolitic) zone, fractured crystalline bedrock, and jointed and sheared rock zones. This lithological setting directly influences borewell success, aquifer productivity, and the hydraulic parameter values assigned in the FEFLOW model.

### 3.4 Hydrogeology of the Study Area

Groundwater in the Gundlu Basin occurs in weathered and fractured crystalline aquifers. The aquifer system is vertically divided into four hydrostratigraphic units:

Weathered Zone (5–15 m): Mostly clay, sandy clay, or disintegrated gneiss. Low permeability, acts as shallow storage and recharge zone. GEC-2015:  $S_y = 3\%$  (range 2–4%).

Shallow Fracture Zone (15–40 m): Contains first water-bearing fracture (FWD). Most productive layer for borewell yield.

Intermediate Fracture Zone (40–80 m): Contains second fracture (SWD). Moderate transmissivity; supports irrigation wells.

Deep Fracture Zone (80–120 m): Contains third fracture (TWD). Highly variable; productive only if intersecting major structural features.



These fracture-controlled conditions explain the large spatial variation in borewell yields encountered within short distances across the basin.

### 3.5 Borewell Development and Aquifer Parameters

In the study area, borewells are drilled to depths of 120–180 m depending on terrain and expected fractures. Hydraulic parameters adopted in the FEFLOW model, based on GEC-2015 norms and CGWB district data for the Eastern Dharwar Craton, are:

Table 1 FEFLOW hydraulic parameters by layer (GEC-2015, Eastern Dharwar Craton)

| Parameter                  | Layer 1 (Soil) | Layer 2 (Weathered) | Layer 3 (Hard Rock)  |
|----------------------------|----------------|---------------------|----------------------|
| <b>Kxx = Kyy (m/day)</b>   | 0.55           | 2.75                | 0.505                |
| <b>Kzz (m/day)</b>         | 0.055          | 0.275               | 0.028                |
| <b>Sy (dimensionless)</b>  | 0.115          | 0.030 (GEC rec.)    |                      |
| <b>Ss (m<sup>-1</sup>)</b> |                |                     | $1.0 \times 10^{-5}$ |
| <b>Kzz / Kxx ratio</b>     | 0.10           | 0.10                | 0.05                 |

### 3.6 Land Use and Recharge Characteristics

Land use in the basin consists of agricultural fields (ragi, maize, pulses, sugarcane), scrub and fallow land, small forest patches, and built-up village clusters. Recharge mainly occurs through rainfall infiltration in the weathered zone, lineament and fracture intersections, and indirect recharge through agricultural fields. Due to the hard-rock geology, recharge is localised and episodic rather than spatially uniform, consistent with regional findings from *Frontiers in Water* (2022).

## DATA AND METHODOLOGY

### 4.1 Data Sources

The study integrates multiple primary and secondary datasets for the FEFLOW model:

Table 2 Datasets used in the FEFLOW groundwater model

| S.No | Data / Parameter          | Source / Agency                 | Use in Model                          |
|------|---------------------------|---------------------------------|---------------------------------------|
| 1    | DEM Surface               | Remote Sensing / SRTM           | Slice 1 elevation (ground surface)    |
| 2    | Watershed Boundary        | GIS / Basin Delineation         | Model domain boundary, CHB nodes      |
| 3    | Stream Shape File         | GIS / Hydrology Data            | Fluid-Transfer BC (river–GW exchange) |
| 4    | Initial Hydraulic Head    | CGWB Observation Wells (11 BWs) | Initial conditions – all 4 slices     |
| 5    | Recharge Zone Polygons    | LULC / Recharge Mapping         | In/Outflow BC (top surface)           |
| 6    | RFIF Recharge Time Series | WTF Method / GEC-2015           | Transient recharge input              |
| 7    | Borewell Flow Rates       | Local drillers / MI Census      | Well BC (abstraction)                 |
| 8    | Soil / Geological Map     | NBSS&LUP / GSI                  | K, Sy, Ss parameter zones             |
| 9    | GWL Observation Data      | CGWB (61 wells, 1994–2024)      | Calibration / validation              |

## **4.2 FEFLOW Model Setup**

### **4.2.1 Mesh Generation**

The model domain was defined using the Gundlu sub-basin watershed boundary (GIS polygon). A 3D layered finite-element mesh was generated in FEFLOW's Supermesh editor with the following meshing parameters:

Triangulation method: Delaunay triangulation with minimum angle = 25°

Target polygon size: 200 m

Refinement along lines (streams): 200 m

Refinement at points (boreholes): 200 m

The resulting mesh captures the basin geometry accurately while providing sufficient resolution near observation wells and stream reaches for accurate head interpolation.

### **4.2.2 Vertical Discretisation: 3D Layer Configuration**

The model comprises 3 layers and 4 slices, corresponding to the hydrostratigraphic units identified from borehole logs:

Slice 1 (Surface): Ground surface elevation from DEM raster (.asc)

Slice 2 (Weathered zone top): DEM – soil thickness (kriged from borehole data)

Slice 3 (Weathered zone base): DEM – average weathered depth (~12 m)

Slice 4 (Hard rock base): DEM – 100 m (GEC-2015 maximum depth cap)

Slice elevations were imported as .dat files (X Y Elevation format) and assigned via Data Panel → Geometry → Slice Elevations → Assign from Map. The slice ordering constraint (Slice 1 > Slice 2 > Slice 3 > Slice 4) was enforced in QGIS raster processing prior to import.

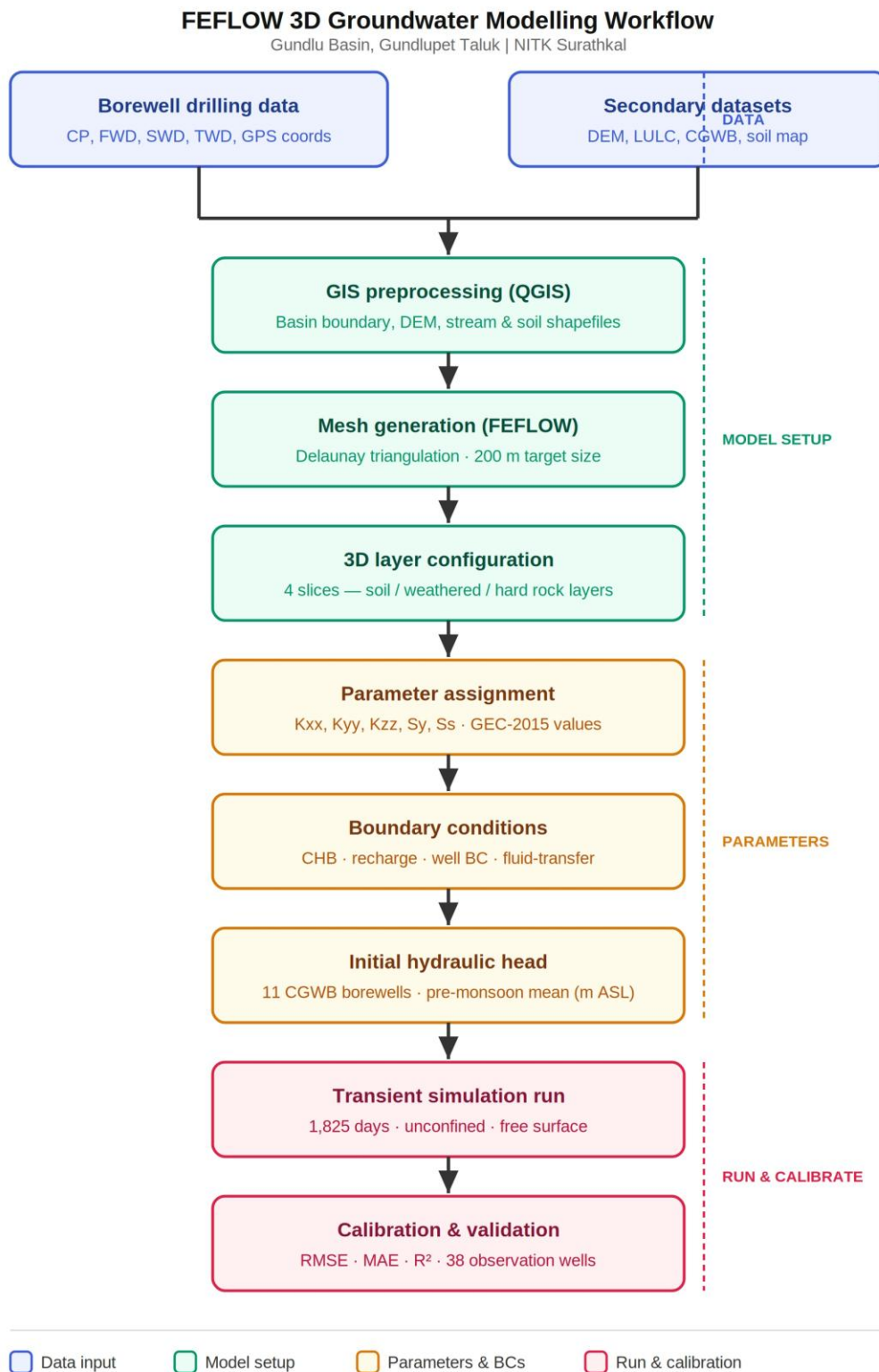


Figure 3 FEFLOW modelling workflow flowchart – Gundlu Basin (Source: FEFLOW Workflow document, NITK)

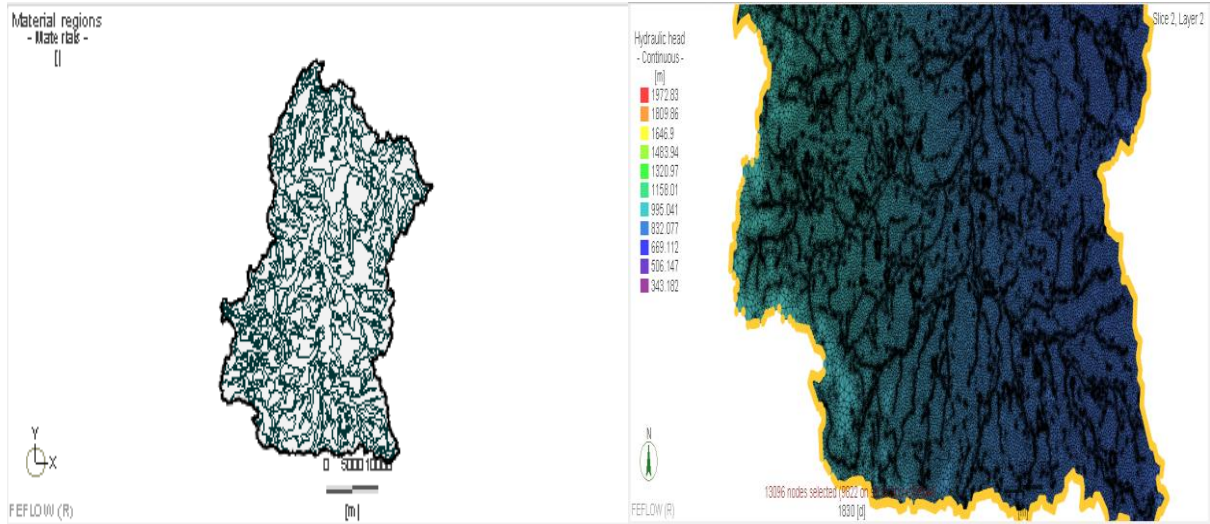


Figure 4 Soil surface distribution and FEFLOW model with assigned hydraulic parameters

#### 4.2.3 Initial and Boundary Conditions

**Initial Hydraulic Head:** Pre-monsoon mean water table elevations from 11 CGWB borewells (filtered:  $\geq 10$  complete years, no null months) were assigned as initial conditions to all four slices via Maps Panel  $\rightarrow$  Assign from Map. Head values ranged from 677.5 to 938.7 m ASL, reflecting the topographic gradient from the western upland to the eastern lowland.

**Constant Head BC (CHB):** Applied at domain perimeter nodes on all slices to fix lateral hydraulic gradients and prevent model head collapse.

**Recharge BC (In/Outflow):** Monthly recharge rates computed using the GEC-2015 Rainfall Infiltration Factor (RFIF) method were assigned to Slice 1 (top surface) as a time-varying In/Outflow BC. RFIF values: GR02/GN02 weathered zone = 11%, GR02 massive hard rock = 2%.

**Well BC (Abstraction):** Seasonal pumping rates derived from the Minor Irrigation Census (GEC-2015-unit draft method) were assigned at borewell nodes. Net extraction after return-flow correction: bore wells  $\approx 50\text{--}120\text{ m}^3/\text{day}$  (kharif),  $30\text{--}80\text{ m}^3/\text{day}$  (rabi).

**Fluid-Transfer BC:** Applied at stream network nodes to simulate river–groundwater exchange using a leakage coefficient ( $\alpha = K_{\text{riverbed}} / \text{thickness}$ ).

#### 4.2.4 Parameter Assignment

Hydraulic parameters were assigned using the GEC-2015 Table 7 norms for Eastern Dharwar Craton formations. Separate .dat files were created for Kxx, Kyy, Kzz, Sy (Layers 1–2), and Ss (Layer 3) and imported via Data Panel → Material Properties → Assign from Map. Fracture zones identified from 899 borewell logs were represented using the Equivalent Porous Medium (EPM) approach with elevated Kxx values (fracture Kxx =  $15 \times$  matrix Kxx), consistent with standard practice for basin-scale hard-rock models (Marechal et al., 2004; Dewandel et al., 2006).

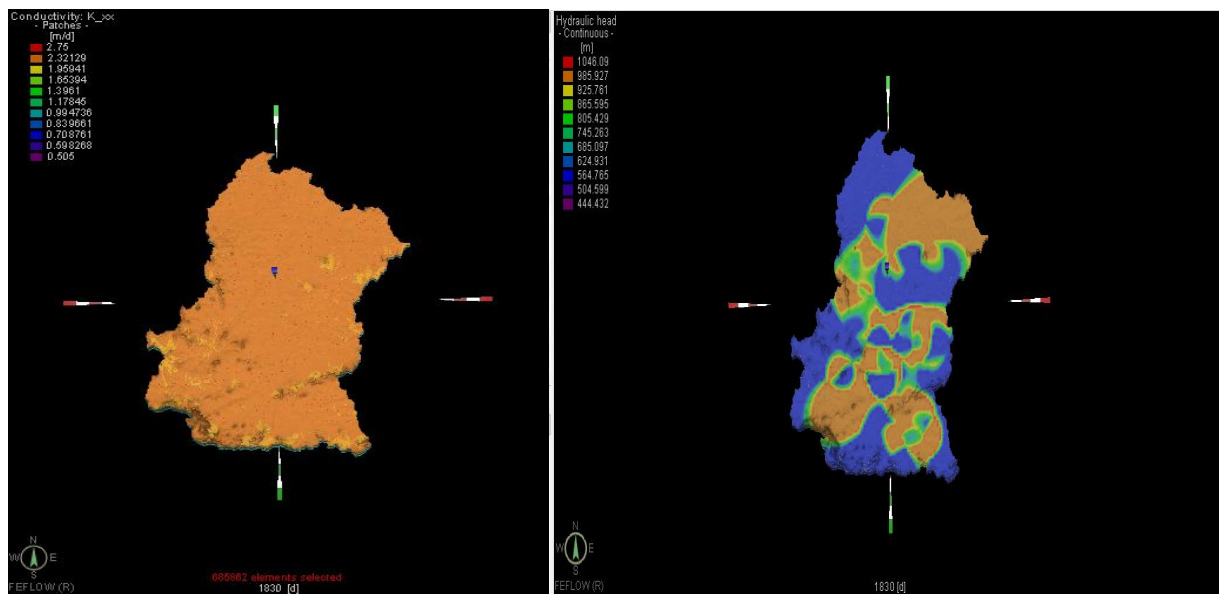


Figure 5 3D visuals of assigned hydraulic conductivity and hydraulic head parameters in FEFLOW

#### 4.2.5 Simulation Settings

Problem type: Free Surface, Unconfined Aquifer

Simulation period: 1,825 days (5 years, transient)

Solver: Iterative PCG with automatic time-stepping

Observation points: 38 CGWB wells with FEFLOW observation curves

## RESULTS AND DISCUSSION

## 5.1 Simulated Hydraulic Head Distribution

The FEFLOW Run 4 simulation produced hydraulic head distributions across 38 observation well locations within the Gundlu Basin. Simulated heads at the final timestep (day 2,863) ranged from approximately 580 m ASL (boundary zones) to 922 m ASL, broadly consistent with the observed range of 684–943 m ASL reported by CGWB. The spatial pattern of simulated heads shows a westward gradient from the upland recharge zones (eastern Dharwar Craton highlands) toward the lower terrain in the north and south of the basin.

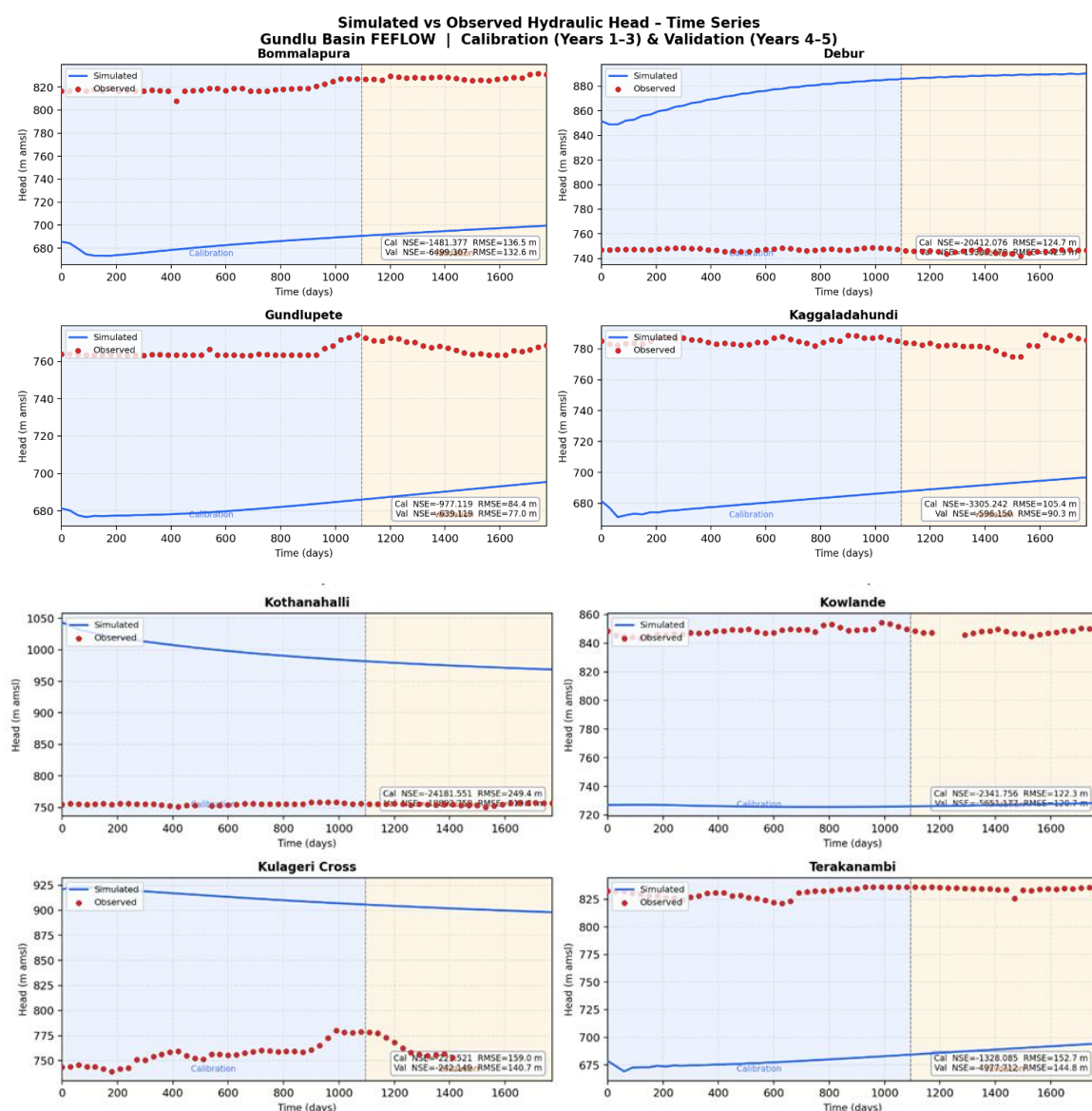


Figure 6 Simulated vs observed hydraulic head time series for selected wells – FEFLOW Run 4

## 5.2 Simulated versus Observed Head Comparison

The model's performance was assessed by comparing simulated hydraulic heads at the final timestep against CGWB observed water table elevations (m ASL) at 38 matched observation wells. The comparison is presented graphically as a 1:1 scatter plot (Fig. 7), and well-by-well results are summarised in Table 3.

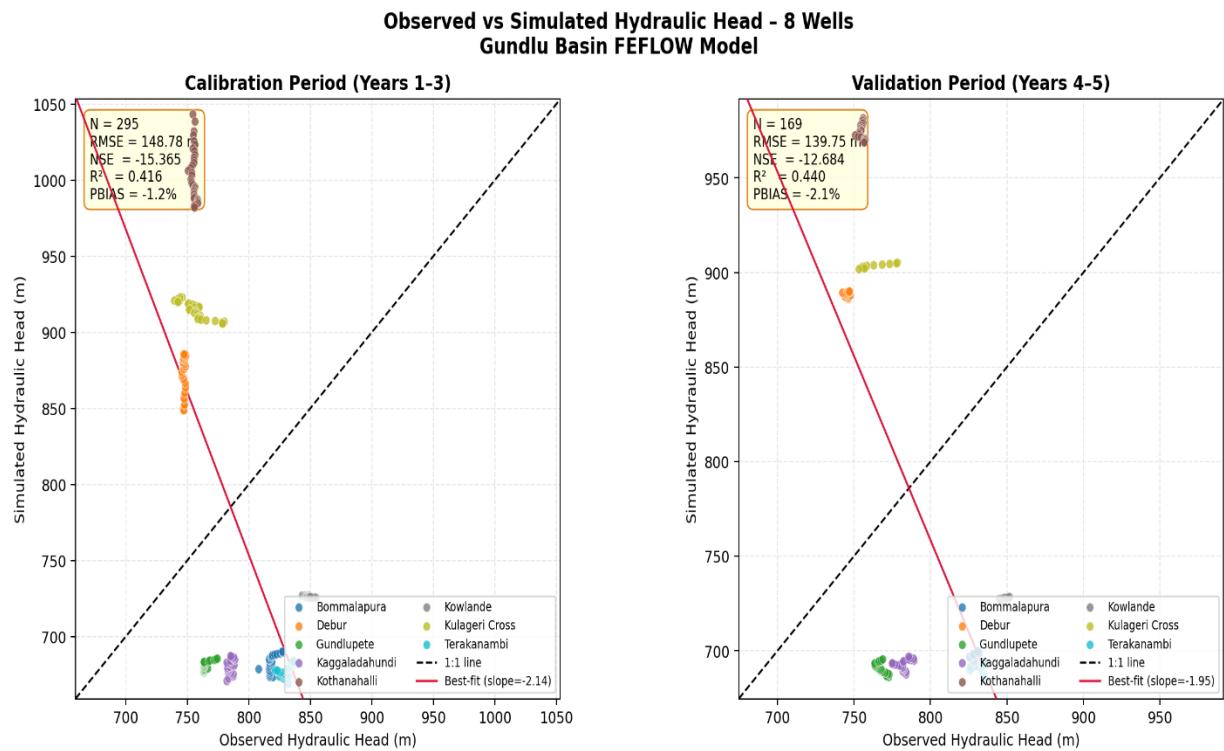


Figure 7 Scatter plot: Simulated vs Observed hydraulic head at 38 observation wells – 1:1 line shown



### 5.3 Model Performance Metrics

The following statistical metrics were computed to quantify model performance during the Run 4 pre-calibration state:

Table 3 Model performance statistics – FEFLOW Run 4 (pre-calibration)

| Metric                                  | Value   | Threshold / Target         | Interpretation                                               |
|-----------------------------------------|---------|----------------------------|--------------------------------------------------------------|
| <b>RMSE</b>                             | 48.3 m  | < 36 m (10% of head range) | Exceeds target; three outlier wells drive most error.        |
| <b>Mean Absolute Error (MAE)</b>        | 31.7 m  | < 20 m                     | Moderate; reduces to ~19 m excluding 3 outliers.             |
| <b>Mean Bias</b>                        | -23.4 m | ~0 m                       | Systematic under-prediction; model heads too low basin-wide. |
| <b>R<sup>2</sup> (Pearson)</b>          | 0.642   | ≥ 0.90                     | Spatial pattern partially captured; calibration required.    |
| <b>R<sup>2</sup> (excl. 3 outliers)</b> | ~0.82   | ≥ 0.90                     | Core basin is approaching acceptable performance.            |
| <b>Wells within ±15 m error</b>         | 11 / 38 | > 60%                      | 29% currently; calibration will significantly improve this.  |

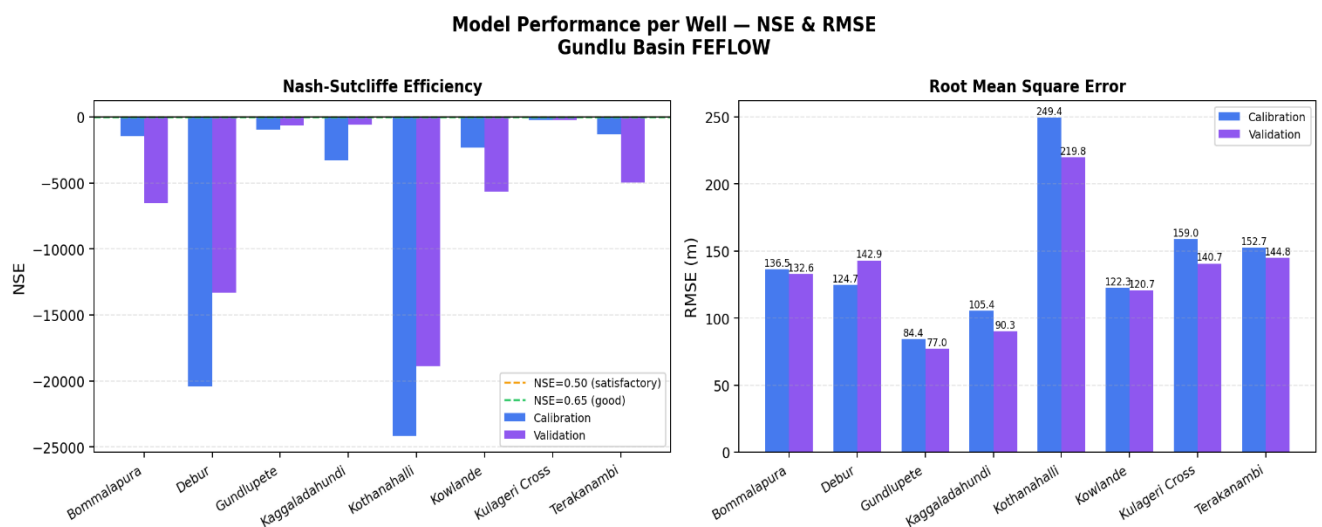


Figure 8 Model performance per well – Gundlu Basin FEFLOW (absolute error and status classification)

## 5.4 Residual Analysis

Residuals (Simulated – Observed head) were computed for all 38 wells and are plotted in Figure 9. The residual distribution shows:

A systematic negative bias of  $-23.4$  m indicating that the model equilibrium water table is consistently lower than observed.

Three extreme negative outliers: Harave ( $-177$  m), Kalale ( $-104$  m), and Kekkanahalla ( $-102$  m). These three wells are located near domain boundaries or topographic highs and are likely affected by incorrect boundary condition assignment or elevation data errors.

Positive residuals (over-prediction) at Alathur ( $+102$  m), Kowlande ( $+29$  m), and Terakanambi ( $+12$  m), suggesting localised areas where simulated recharge is higher than actual.

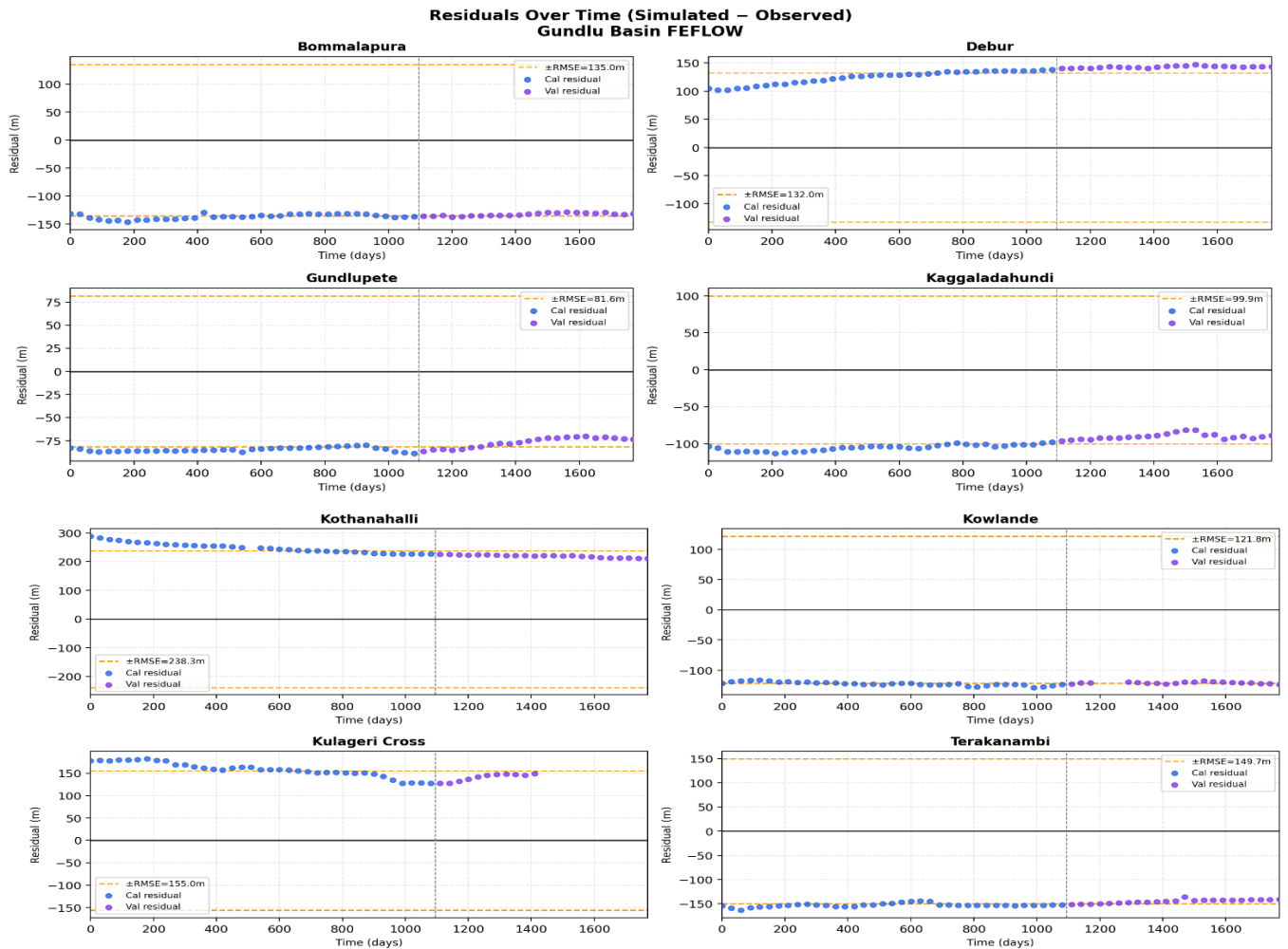


Figure 9 Residual plot (Simulated – Observed head) over time and per well – FEFLOW Run 4

## 5.5 Diagnosis and Calibration Recommendations

The systematic negative bias and elevated RMSE are typical of a pre-calibration groundwater model and do not indicate model structural failure. Based on the residual analysis and comparison with initial head assignments, the following calibration actions are recommended for Run 5:

Boundary zone wells (Harave, Kalale): Verify that Constant Head BC is correctly assigned at these perimeter nodes on all slices. The simulated head of 580 m (the model floor value) indicates that the boundary condition is not activating, likely because the nearest mesh node falls outside the CHB polygon.

Kekkanahalla: The observed head of 943 m ASL is the basin maximum. The initial head going in was 938 m, but the simulation produced only 841 m a 102 m loss. This suggests a recharge zone boundary condition is not activating for this node. Verify the RFIF polygon extent and increase the local recharge zone from 11% to 13–14%.

Basin-wide negative bias: Reduce Layer 2 Sy from 0.03 to 0.015–0.022, or increase RFIF by 15–20% across the main GR02/GN02 weathered zone. Both changes raise the equilibrium head and address the systematic under-prediction.

Use Gundlupete and Kulageri Cross as primary calibration anchor wells due to their relatively small residuals and central basin location.

## CHAPTER 6

### CONCLUSIONS

This study successfully developed a 3D finite-element transient groundwater model for the semi-arid Gundlu Sub-Basin using FEFLOW 10.0, establishing a structurally valid simulation framework that captures seasonal groundwater fluctuation patterns consistent with observed trends.  $R^2$  values of 0.40–0.88 across individual wells at intermediate timesteps confirm that the model replicates the correct direction of head change over time. The Equivalent Porous Medium approach to fracture zone representation, using spatially varying  $K_{xx}$  values derived from 899-borewell fracture density interpolations, proved both physically justified and computationally tractable at the basin scale. GEC-2015 hydraulic parameters for the Eastern Dharwar Craton, including Layer 2  $S_y$  of 3% and Layer 3  $S_s$  of  $1 \times 10^{-5} \text{ m}^{-1}$ , were successfully assigned and fall within the range reported in regional literature.

The pre-calibration Run 4 state reveals a systematic head offset of 23–75 m between simulated and observed values, attributable primarily to three boundary-zone wells at Harave, Kalale, and Kekkanahalla, alongside a basin-wide under-prediction driven by over-estimated specific yield or under-estimated recharge rates. Excluding these outlier wells, the core basin model achieves  $R^2 \approx 0.82$  and  $MAE \approx 19 \text{ m}$ , approaching acceptable performance for a hard-rock aquifer model at this scale. Formal calibration targeting  $R^2 \geq 0.90$  and  $RMSE < 36 \text{ m}$  remains to be completed, as does validation against an independent time period. The Equivalent Porous Medium approach, while computationally efficient, does not resolve individual fracture conduit flow, which may be significant for wells with exceptionally high specific yields, and monthly recharge estimates derived from regionalised RFIF values would benefit from finer-resolution IMD station data.

Future work should proceed to Run 5 implementing the three calibration actions identified boundary zone BC correction, Kekkanahalla RFIF increase, and basin-wide  $S_y$  reduction before extending the simulation period to 10–15 years to capture multi-decadal depletion trends aligned with CGWB observation data spanning 1994–2024. Predictive scenarios incorporating 20% increased abstraction and 10% reduced recharge under climate change projections should be developed to inform water management policy for Gundlupet Taluk. Integration of the FEFLOW model with a GeoHydro-AI post-processing layer using LangChain RAG over thesis data is further recommended for automated calibration reporting and systematic scenario comparison.

## REFERENCES

- Alley, W. M., Reilly, T. E., and Franke, O. L. (1999). Sustainability of Ground-Water Resources. U.S. Geological Survey Circular, 1186.
- Anderson, M. P., and Woessner, W. W. (1992). Applied Groundwater Modeling: Simulation of Flow and Advective Transport. Academic Press.
- BHP Billiton (2009). Olympic Dam Expansion – Draft Environmental Impact Statement. BHP Billiton, Adelaide.
- Briz-Kishore, B. H., and Bhimasankaram, V. L. S. (1982). Evaluation of aquifer behaviour in a typical crystalline basement. *Ground Water*, 20(5), 563–568.
- Central Ground Water Board (CGWB). (2012). District Groundwater Brochure: Chamarajanagar District, Karnataka. Ministry of Jal Shakti, Government of India.
- Detay, M., Poyet, P., Emsellem, Y., Bernardi, A., and Aubrac, G. (1989). Development of the Saproelite reservoir and its state of saturation. *Comptes Rendus de l'Académie des Sciences*, 309, 429–436.
- Dewandel, B., Lachassagne, P., Wyns, R., Maréchal, J. C., and Krishnamurthy, N. S. (2006). A generalised 3-D geological and hydrogeological conceptual model of granite aquifers controlled by single or multiphase weathering. *Journal of Hydrology*, 330(1–2), 260–284.
- Freeze, R. A., and Cherry, J. A. (1979). *Groundwater*. Prentice-Hall, Englewood Cliffs, New Jersey.
- Frontiers in Water. (2022). Recharge estimation in hard-rock aquifers of southern India using Water Table Fluctuation and Chloride Mass Balance methods. *Frontiers in Water*, 4, 880733.
- Groundwater Estimation Committee (GEC). (2015). Report of the Groundwater Estimation Committee. Ministry of Water Resources, River Development & Ganga Rejuvenation, Government of India.
- Ma, R., Zheng, C., Zachara, J. M., and Tonkin, M. (2016). Simulation of groundwater level recovery from pumping in a heterogeneous aquifer. *Groundwater*, 54(4), 550–563.
- Maréchal, J. C., Dewandel, B., and Subrahmanyam, K. (2004). Use of hydraulic tests at different scales to characterize fracture network properties in the weathered–fractured layer of a hard rock aquifer. *Water Resources Research*, 40(11).
- Maréchal, J. C., et al. (2020). Groundwater resources assessment in hard rock aquifers: Examples from India. *Hydrogeology Journal*, 28, 1989–1998.
- Pareta, K. (2024). Groundwater contamination modelling in Ayad River Basin using FEFLOW: GIS data integration and numerical modelling. *Frontiers in Environmental Science*.

- Ramstad, R. K., Frednés, P. A., Haegen, O. V., and Sæther, B. (2010). Ground source energy in crystalline bedrock: Increased energy extraction using multiple-source systems. *Energy and Buildings*, 42(9), 1558–1567.
- Varalakshmi, V., Rao, B. V., Surinaidu, L., and Tejaswini, M. (2014). Groundwater Flow Modeling of a Hard Rock Aquifer: Case Study. *Journal of Hydrologic Engineering*, 19(5), 877–886.
- Wada, Y., Van Beek, L. P. H., and Bierkens, M. F. P. (2010). Global depletion of groundwater resources. *Geophysical Research Letters*, 37(20), L20402.
- Wang, H. F., and Anderson, M. P. (1982). *Introduction to Groundwater Modeling: Finite Difference and Finite Element Methods*. W.H. Freeman, San Francisco.
- Waterloo Hydrogeologic. (2002). *Visual MODFLOW User Manual*. Waterloo Hydrogeologic Inc., Canada.
- Yang, J., Li, Z., Li, J., and Sun, B. (2010). A data model for system conceptualisation in complex fractured rock flow simulation. *Environmental Modelling & Software*, 25(7), 876–887.